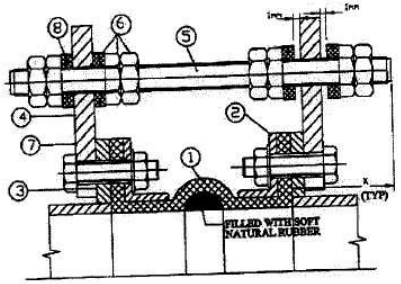
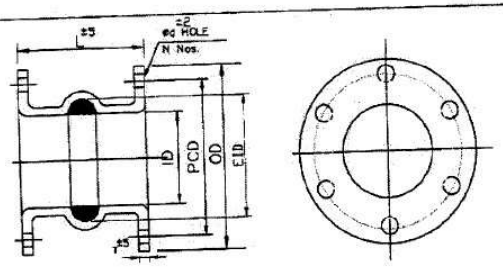
ALL DIMENSIONS ARE IN "MM" UNLESS OTHERWISE SPECIFIED



RUBBER EXPANSION JOINT ASSY

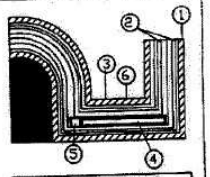
2000NB - (TIED)



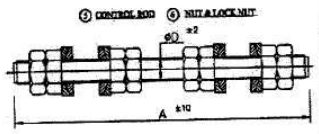
① RUBBER EXPANSION JOINT

TOLERANCE: AS PER FSA STANDARD (PAGE NO.19) (SHEET ENCLOSED) * EFFECTIVE ID

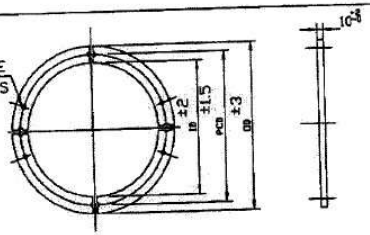
SNo	ID	PCD	OD	øD	N	L	T	E ID	X
1	2000	2315.7	2419.35	54	64	350	35	2180	195



CROSS SECTION DETAIL (NOMINATE)	
1	INNER TUBE: NITRILE(4.5 thk)
2	NYLON FABRIC HIGH STRENGH
3	FILLER RUBBER (NATURAL)
4	RECTANGULAR BODY RING (CS) ASTM A517 Gr.F
5	SQUARE ROD (EN 8)
6	OUTER COVER:NEOPRENE(4.5 thk)

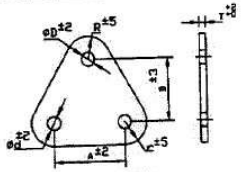


S.No	A	øD	QTY/REJ.
1	740	39	8



② RETAINER RING
(SPLIT INTO 4 SEGMENTS)

SNo	NB	ID	PCD	OD	ød	N
1	2000	2120	2315.7	2419.35	54	64



④ STRETCHER PLATE

SNo	NB	A	B	øD	ød	R	r	T	QTY/REJ.
1	2000	114	115	42	54	40	55	45	16

⑦ BOLT NUT & WASHER

BELLOW ID	ON STRETCHER PLATE SIDE SIZE	QTY/R.E.J	ON C.FLANGE SIDE SIZE	QTY/R.E.J
2000	M48 x 210 ±5	32	M48 x 170 ±5	96

STIFFNESS VALUE(N/MM)

SIZE	LATERAL	AXIAL
2800 NB	3254	2900
2000 NB	2254	2030

NOTE: THE ABOVE VALUES ARE BASED ON 'O' PRESSURE CONDITION AT ROOM TEMP.

POS	DESCRIPTION	QTY	MATERIAL
8	SPHERICAL WASHER (DOUBLE DIP E-GALV.)	SEE TABLE	IS: 2062
7	BOLT,NUT&WASHER (MS) (DOUBLE DIP E-GALV.)		IS: 1367 CL:8.8/8.0
6	NUT&LOCK NUT (DOUBLE DIP E-GALV.)		IS 1367 Cl 8.0
5	CONTROL ROD (DOUBLE DIP E-GALV.)		IS 1367 Cl 8.8
4	STRETCHER PLATE (DOUBLE DIP E-GALV.)		ASTM A 516 Gr.60/70
2	RETAINER RING (DOUBLE DIP E-GALV.)	2	ASTM A 516 Gr.60/70
1	RUBBER EXPANSION JOINTS	1	RUBBER

Tata Power Mundra

SHROFF PROCESS PUMPS

Title: G.A DRAWING OF RUBBER EXPANSION JOINT

DESIGN DATA :-

MATERIAL	: NITRILE INNER TUBE WITH NEOPRENE OUTER COVER
DESIGN PRESSURE	: 5.0 Kg/Sq.CM
HYDRO TEST PRESSURE	: 7.5 Kg/Sq.Cm (g) at room temp.
TEST VACUUM	: 700 mm of Hg
FLOWING MEDIUM	: SEA WATER
SHORE HARDNESS	: 65±5° Sh. A
TEMPERATURE	: MAX 50° C
DRILLING STD	: AWWA C 207 CL.B FOR 2000NB As per Annexure-III of Technical Contract Document for 2800 NB
PRESSURE DROP ACROSS THE EXP.JOINT	: 0.01 MWC

MOVEMENTS AT DESIGN PRESSURE

AXIAL EXPANSION	FOR S.NO:1&2	: 0 mm
AXIAL COMPRESSION		: 0 mm
AXIAL EXPANSION	FOR S.NO:3	: 1 mm
AXIAL COMPRESSION		: 1 mm
LATERAL DEFLECTION		: 12 mm
ANGULAR DEFLECTION		: 1°

PAINTING

TWO COATS OF ZINC PHOSPHATE RICH PRIMER (EPOXY BASED) OF THICKNESS 80u AND TWO COATS OF CHLORINTED RUBBER OF THICKNESS OF 80u FOR INDOOR AND 100u FOR OUTER LOCATION.